CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rhombus has perimeter 340 cm and one diagonal 26 cm. Its area (in cm²) is: 1

(a) 1092 **(b)** 2184 **(c)** 2210 **(d)** 2197

2. A ladder 17 m long, standing in a street, rests against a house on one side when its top reaches 15 m. Without moving its foot it is turned to rest against the opposite house, reaching 8 m. The width of the street (in m) is: 1

(a) 24 **(b)** 17 **(c)** 23 **(d)** 25

3. A square has a diagonal of $8\sqrt{2}$ cm. Its area (in cm^2) is: 1

(a) 128 **(b)** 32 **(c)** 64 **(d)** 256

4. A ladder 85 m long reaches 77 m up a wall. Its top slides down 2 m along the wall. The foot of the ladder moves away from the wall by (in m): 1

(a) 2 **(b)** 4 **(c)** 40 **(d)** 36

5. The legs of a right triangle are in the ratio $7 : 24$ and its hypotenuse is 50 cm. Its perimeter (in cm) is: 1

(a) 62 **(b)** 114 **(c)** 112 **(d)** 100

6. The sides of a rectangle are in the ratio $8 : 15$ and its diagonal is 34 cm. The area of the rectangle (in cm²) is: 1

(a) 92 **(b)** 68 **(c)** 484 **(d)** 480

7. A Pythagorean triplet whose smallest member is 22 is: 1

(a) 22, 121, 123 **(b)** 22, 120, 122 **(c)** 22, 119, 121 **(d)** 22, 23, 24

8. A rhombus has perimeter 340 cm and one diagonal 72 cm. Its area (in cm²) is: 1

(a) 6120 **(b)** 2772 **(c)** 5544 **(d)** 5580

9. **Assertion (A):** In a right-angled triangle the hypotenuse is the longest side. 1
Reason (R): In a right-angled triangle the square of any side equals the sum of the squares of the other two sides.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The diagonal of a square of side 5 cm is 10 cm. 1
Reason (R): The diagonal of a square of side a is $a\sqrt{2}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find a Pythagorean triplet in which one member is 24. 2

12. A kite string 65 m long is stretched straight and the kite is 63 m above the boy's hand. Find the horizontal distance of the boy from the point directly below the kite. 2

13. The legs of a right triangle are in the ratio $8 : 15$ and the hypotenuse is 68 cm. Find the legs. 2

14. Find a Pythagorean triplet in which one member is 33. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 65 m long, with its foot in a street, reaches a window 63 m high on one side. Without moving its foot it is turned to reach a window 33 m high on the opposite side. Find the width of the street. 3

16. A rhombus has perimeter 340 cm and one diagonal 72 cm. Find the other diagonal and the area. 3

17. Two ships leave a port, one sailing east at 20 km/h and the other south at 21 km/h. How far apart are they after 4 hours? 3

OR

A ladder 61 m long, with its foot in a street, reaches a window 60 m high on one side. Without moving its foot it is turned to reach a window 11 m high on the opposite side. Find the width of the street. 3

18. A rhombus has perimeter 356 cm and one diagonal 78 cm. Find the other diagonal and the area. 3

OR

Two ships leave a port, one sailing east at 9 km/h and the other south at 40 km/h. How far apart are they after 2 hours? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Raju must go from corner A to the opposite corner C of a rectangular field 40 m long and 30 m wide, either along the two sides or straight across along the diagonal. 5

(i) Find the length of the diagonal.
(ii) Find the distance along the two sides.
(iii) How much distance does he save by walking across?
(iv) Kavya says $(15, 20, 25)$ is also a Pythagorean triplet because it is half of the field's triplet. Verify her statement.

20. In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 12 cm, BC = 16 cm, CD = 21 cm and DA = 29 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

OR

A ladder 65 m long reaches a point 60 m up a wall. Its top slides down 8 m. Find 5

(i) the original distance of the foot from the wall,
(ii) its new distance from the wall,
(iii) how far the foot moves.

END OF QUESTION PAPER — Total: 40 Marks